Off-market prediction

90-second speaker script · Member B

190 written words. Rehearse at about 130 words per minute, with brief pauses at slide changes.

Slide 1 — Working process | 0:00–0:48

Here is how my off-market model works. First, I keep only information available without an asking price. I turn area ranges into midpoints, create property ratios, and extract fixed description cues. Category encoding is fitted inside each training fold, and encoded categories remain categorical. Histogram gradient boosting groups numeric measurements into bins. It also learns how to handle missing measurements. It then adds small decision trees that correct remaining errors in log sale price. Finally, it combines their contributions and converts the prediction back into dollars. This captures nonlinear relationships while keeping list price and post-sale information out of the predictors.

Slide 2 — Validation and results | 0:48–1:10

On the graph, median absolute percentage error drops from twenty point eight four percent for the nearest-sales reference to eight point five three percent. Mean absolute percentage error is eleven percent. We tested ten thousand and sixty-nine homes across six monthly folds, training only on earlier sales.

Slide 3 — Refit, predict, and check | 1:10–1:30

The monthly graph checks performance over time. After validation, I refitted on all one hundred and fifty thousand original sales and produced three thousand nine hundred and seventy-four predictions. Every test ID appears once, with positive finite prices. Hidden-test accuracy remains unknown.

Timing note: your 90 seconds leaves 3 minutes 30 seconds for the rest of the five-minute team video.